

Sejal Bhatia

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EXPERIENCE

Ochsner Health

Machine Learning Engineer

Jan 2025 - Present

- Designed and built a robust data labeling system for clinicians to annotate nasal endoscopic data, leveraging **FastAPI**, **PostgreSQL**, and integrated tools like **COCO and CVAT Annotator**.
- Developed an AI model for mucus segmentation in sinusitis diagnosis, achieving near-human performance (71% Dice, 63% IOU), leveraging **Swin-UNet**, adaptive convolutions, and attention mechanisms to tackle thin mucus detection, boundary precision, and illumination artifacts.
- **Deployed** AI models on **NVIDIA Jetson Orin Nano edge devices** for real-time inference during nasal endoscopy; enabled remote management and updates using **Kubernetes and KubeEdge**, leveraging MQTT for efficient device communication and scalable deployment.
- **Directed a team to upgrade our annotation backend**, migrating all annotations to CVAT and adding support for advanced search, versioning, YOLO export, and SLAM/tracking workflows.
- Contributed to 4 research paper submissions at international conferences (MICCAI, AMIA, IFAR) and delivered 5 poster and podium presentations at Ochsner Research Week.
- Architected and maintained **end-to-end ML pipelines** for medical image analysis, including data preprocessing, **multi-GPU training** on on-premise servers, and optimized model deployment; leveraged **CUDA, PyTorch**, and performance tuning techniques for real-time inference and system reliability.
- Generated propagated annotations via **2D tracking, 3D SLAM, and CT back-projection**; built pipelines for ingesting and validating them in CVAT and Catalog.
- **Optimized PostgreSQL performance** for high-volume data ingestion through schema normalization, strategic indexing, and Alembic migrations for version control

ITArks Inc

Software Developer Intern

Aug 2024 - Dec 2024

- Built a computer vision pipeline for early diabetic retinopathy screening using CLAHE-based contrast enhancement, morphological filtering, and U-Net variants, supporting
- Preprocessed fundus images using OpenCV and scikit-image to evaluate brightness, blur (Laplacian variance), and contrast entropy, reducing dataset noise by 35%
- Developed a FastAPI-based REST service to expose lesion segmentation predictions, supporting asynchronous uploads, report generation, and seamless integration with an ophthalmology platform
- Conducted retrospective validation on IDRiD and APTOS datasets, achieving F1-score greater than 0.81 and AUC of 0.89 on microaneurysm detection
- Ensured HIPAA compliance through automated DICOM anonymization and secure AWS S3-based storage pipelines.
- Authored technical documentation and API specifications for handoff to a clinical engineering team, participating in design reviews to ensure secure deployment

University of Colorado

Graduate Assistant for University Advising

Feb 2023 - May 2024

- Developed Python-based ETL pipelines to automate ingestion and cleaning of student records from multiple databases and flat files, improving data accessibility and reducing manual work by 60%.
- Built predictive models using scikit-learn to forecast advising workloads and dropout risks based on academic history, engagement metrics, and demographics.
- Designed interactive Tableau dashboards and Matplotlib visualizations to present insights on advising trends and student success to university leadership.
- Applied SQL joins, window functions, and aggregation queries to extract and transform data from relational systems; optimized queries for performance.

EDUCATION

Master of Science in Computer Science, University of Colorado

Aug 2022 - May 2024

Bachelor of Engineering in Computer Science and Engineering, Pune University, India

Aug 2018 - May 2022

SKILLS

Programming Languages	Python, Java, C, C++, JavaScript, TypeScript, Golang, SQL, Shell Scripting, HTML, CSS
ML-Ops & Dev-Ops Tools	Docker, Kubernetes, MLflow, Jenkins, Argo CD, Helm Charts, Terraform, Ansible, Prometheus, Grafana, AWS Step Functions
Cloud & Operating Systems	Linux, Windows, Microsoft Azure, Google Cloud Platform (GCP), Amazon Web Services (AWS)
Web Development Frameworks	Node JS, React JS, Angular JS, Express JS, Flask, Django, Fast API
ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, TorchServe, ONNX, TensorRT
Data Science Frameworks	Numpy, Pandas, PySpark, Matplotlib, Tableau, PowerBI
Databases	MySQL, Mongo DB, Redis, PostgreSQL
Bigdata Technologies	Kafka, Hadoop, Spark, Snowflake, Elastic Search

PROJECTS

Smart Activity Monitoring: Insights from Wearable Sensor Data

- Developed a machine learning model using Python libraries such as XGBoost, LightGBM, and CatBoost to classify physical activities with 97.85% accuracy. Enhanced performance through sophisticated data preprocessing, feature engineering, MLOps practices, and real-time data processing with Kafka.

Signature Similarity Detection using OpenCV

- Created a signature verification system with OpenCV and Python, using image processing techniques like Canny Edge Detection and Gaussian Blur, resulting in a high-precision comparison algorithm.

Not playing Kirby's Dream Land as intended with Reinforcement Learning

- Developed an AI for "Kirby's Dream Land" using Python, PyTorch, and PyBoy, leveraging Q-learning and DDQN in an OpenAI Gym environment to optimize in-game performance through reinforcement learning strategies.

Dynamic Pricing of food products to optimize supply chain management using machine learning

- Developed a dynamic pricing model for perishable goods using Python for regression analysis and the Bellman equation, leveraging datasets from Kaggle to implement Linear and Polynomial Regression models, reducing food wastage and boosting sales.

Multilingual Text and Speech Translation System

- Created a multilingual translation system for instant speech and text translation using sequence-to-sequence architecture, employing NLP, DSP, and RNN technologies for real-time accuracy.

CERTIFICATIONS & PUBLICATIONS

- [Google Advanced Data Analytics Certificate](#)
- [Google Cloud Data Analytics Certificate](#)
- [Microsoft Azure Data Engineering Associate \(DP-203\) Specialization](#)
- [Activity Identification using Supervised Machine Learning on Wearable Activity Tracker Data \(IEEE\)](#)
- [Application and evaluation of Machine Learning for news article popularity prediction \(IEEE\)](#)
- [Pragmatic segmentation-based credit risk management using Machine Learning \(IEEE\)](#)